Lancaster University Leipzig

Robust Bayes Acts under Prior Perturbations: Contamination, Stability, and Selection Paths

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Outline

Basic Decsion Theory

Maximizing Expected Utility?

Measuring Robustness of Bayes-Acts

Toy Case Study

Basic Decsion Theory

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Informal description of the model:

- An agent has to choose among different acts X from a set $\mathcal{G}$.
- The consequence that choosing X yields depends on which state of nature s from a set S is the true one.

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- Let C denote some non-empty set of consequences.
- Each act X corresponds to a mapping $X : S \to C$.
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Goal: Determining a choice function

$$ch : 2^{\mathcal{G}} \to 2^{\mathcal{G}} \quad \text{with} \quad ch(\mathcal{D}) \subseteq \mathcal{D} \quad \text{for all} \quad \mathcal{D} \in 2^{\mathcal{G}}$$

that best possibly utilizes the available information.

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Strong interpretation:

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Weak interpretation:

- $ch(\mathcal{D})$ is the set of all non-neglectable acts from $\mathcal{D}$.
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Weak interpretation:

- $ch(\mathcal{D})$ is the set of all non-neglectable acts from $\mathcal{D}$.
- These acts are incomparable for the agent.

Obvious comment: If only weakly structured information is available, we often have to work with weakly interpretable choice functions.

Maximizing Expected Utility?

Finite Cardinal Decision Problem

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Assume we have:

- Acts $\mathcal{G} = \{X_1, \ldots, X_n\}$
- States $S = \{s_1, \ldots, s_m\}$
- A cardinal utility function $u : C \to \mathbb{R}$

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A convenient form or analyzing these problems are utility tables:

	s_1	s_2	$\ldots\ldots$	s_{m-1}	s_m
X_1	u_{11}	u_{12}	$\ldots\ldots$	u_{1m-1}	u_{1m}
$\vdots$	$\ldots$	$\ldots$	$\ldots\ldots$	$\ldots$	$\ldots$
X_n	u_{n1}	u_{n2}	$\ldots\ldots$	u_{nm-1}	u_{nm}

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Formally, we have the following criterion:

Bayes-Criterion (a.k.a. Maximizing Expected Utility):

Define the function

$$\Phi_{\texttt{Bayes}} : \mathcal{G} \to \mathbb{R} \quad , \quad X \mapsto \mathbb{E}_\pi(u \circ X).$$

Then, for all $\mathcal{D} \subseteq \mathcal{G}$, we set $ch_\pi(\mathcal{D}) := \operatorname{argmax}_{\mathcal{D}} \Phi_{\texttt{Bayes}}(X)$.

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Regularisation approach: Fix trust parameter $\mu \in [0, 1]$.

Define

$$\Phi_{\mathtt{H\&L}} : \mathcal{G} \to \mathbb{R} \quad , \quad X \mapsto \mu \cdot \inf_{s \in S} u \circ X(s) + (1 - \mu) \cdot \mathbb{E}_\pi(u \circ X)$$

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Imprecise probabilistic approach: Fix credal set $\mathcal{M}$ of candidate probabilities.

Define

$$\Phi_{\mathcal{M}} : \mathcal{G} \to \mathbb{R} \quad , \quad X \mapsto f(\{\mathbb{E}_\pi[u \circ X] : \pi \in \mathcal{M}\}),$$

where $f : 2^{\mathbb{R}} \to \mathbb{R}$ is an aggregation function.

Then, for all $\mathcal{D} \subseteq \mathcal{G}$, we set $ch_{\mathcal{M}}(\mathcal{D}) := \text{argmax}_{\mathcal{D}} \Phi_{\mathcal{M}}(X)$.

Measuring Robustness of Bayes-Acts

Measuring Stability of Bayes-Optimal Acts

Consider the following setup:

- Let $(\mathcal{G}, S, u)$ be a finite cardinal decision problem.
- Let Δ_S denote the probability simplex over the state space (S, Σ_S).
- Let $\pi_0 \in \Delta_S$, and let $\mathcal{G}_{\pi_0}$ the set of Bayes-optimal acts from $\mathcal{G}$ wrt π.
- For $\varepsilon \in [0, 1]$ fixed, let

$$B(\pi_0, \varepsilon) := \Big\{ \pi \in \Delta_S : |\pi_0(s_j) - \pi(s_j)| \leq \varepsilon \text{ for all } j = 1, \ldots, m \Big\}$$

How much do you trust your prior choice π_0?

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$$B(\pi_0, \varepsilon) := \{\pi \in \Delta_S : |\pi_0(s_j) - \pi_0(s_j)| \leq \varepsilon \text{ for all } j = 1, \ldots, m\}$$

Fix an act $X \in \mathcal{G}$, and define the following two quantities:

i) The **robustness radius** of X with respect to π_0 is defined by:

$$rob(X, \pi_0) := \sup \left\{\varepsilon : X \in \mathcal{G}_\pi \text{ for all } \pi \in B(\pi_0, \varepsilon)\right\}$$

ii) The **contamination need** of X with respect to π_0 is defined by

$$con(X, \pi_0) := \inf \left\{\varepsilon : X \in \mathcal{G}_\pi \text{ for some } \pi \in B(\pi_0, \varepsilon)\right\}$$

Intuition behind the formulas:

- $rob(X, \pi_0)$ measures how dependent the optimality of X is on the correct specification of π_0.
- $con(X, \pi_0)$ measures how dependent the sub-optimality of X is on the correct specification of π_0.

Computing The Contamination Need

In the situation before, the contamination need of X with respect to π_0 can be computed via the following linear program:

$$con(X,\pi_0) = \min_{\pi,\varepsilon} \left\{ \varepsilon : \begin{array}{l} \forall j : \pi_j \geq 0 \\ \sum_{j=1}^{m} \pi_j = 1 \\ \forall j : \pi_j \leq \pi_{0j} + \varepsilon \\ \forall j : \pi_{0j} - \varepsilon \leq \pi_j \\ \forall Y \neq X : \sum_{j=1}^{m} \pi_j \big(u \circ X(s_j) - u \circ Y(s_j) \big) \geq 0 \end{array} \right\}$$

Computing The Robustness Radius

In the situation before, fix $\varepsilon \in [0,1]$, $X \in \mathcal{G}_{\pi_0}$ and $Y \neq X$.

Consider the LP:

$$R_{X,Y}(\varepsilon) = \min_{\pi} \left\{ \sum_{j=1}^{m} \pi_j \big(u \circ X(s_j) - u \circ Y(s_j) \big) : \begin{array}{l} \forall j : \pi_j \geq 0 \\ \sum_{j=1}^{m} \pi_j = 1 \\ \forall j : \pi_j \leq \pi_{0j} + \varepsilon \\ \forall j : \pi_{0j} - \varepsilon \leq \pi_j \end{array} \right\}$$

Then, setting $R(\varepsilon) := \min_{Y \neq X} R_{X,Y}(\varepsilon)$, the robustness radius of X with respect to π_0 can be computed via:

$$\mathrm{rob}(X, \pi_0) = \sup \{ \varepsilon \geq 0 : R(\varepsilon) \geq 0 \}$$

Computing The Robustness Radius, continued

Input: decision problem $(\mathcal{G}, S, u)$, prior π_0, act $X \in \mathcal{G}_{\pi_0}$, tolerance $\delta > 0$
Initialize: $\varepsilon_L \leftarrow 0$, $\varepsilon_U \leftarrow 1$
Define: $R(\varepsilon)$ as in Proposition above
while $\varepsilon_U - \varepsilon_L > \delta$ **do**
    $\varepsilon \leftarrow (\varepsilon_L + \varepsilon_U)/2$
    Compute $R(\varepsilon)$ by solving $R_{X,Y}(\varepsilon)$ for all $Y \neq X$
    if $R(\varepsilon) \geq 0$ **then**
        $\varepsilon_L \leftarrow \varepsilon$
    else
        $\varepsilon_U \leftarrow \varepsilon$
    end if
end while
return ε_L $=0$

Going from here: A Stability Criterion

Question: Can we design a meaningful decision criterion that balances both stability measures (*rob* and *con*) from before?

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Idea: In the situation before, we fix

- a selection cost function $c : \mathcal{G} \to \mathbb{R}_{\geq 0}$, and
- a weight parameter $\lambda \geq 0$.

Define the **cost-adjusted stability score** by

$$S_{\lambda,\pi_0}(X) := \begin{cases} \frac{\mathrm{rob}(X,\pi_0)}{\max_{Y \in \mathcal{G}} \mathrm{rob}(Y,\pi_0)} - \lambda \frac{c(X)}{\max_{Y \in \mathcal{G}} c(Y)}, & \text{if } X \in \mathcal{G}_{\pi_0}, \\ -\frac{\mathrm{con}(X,\pi_0)}{\max_{Y \in \mathcal{G}} \mathrm{con}(Y,\pi_0)} - \lambda \frac{c(X)}{\max_{Y \in \mathcal{G}} c(Y)}, & \text{if } X \notin \mathcal{G}_{\pi_0} \end{cases}$$

A **cost-adjusted stability optimal act** is any

$$X^* \in \arg\max_{X \in \mathcal{G}} S_{\lambda,\pi_0}(X).$$

Toy Case Study

A Toy Case Study: Portfolio Selection

Situation:

- We use financial return data to approximate a decision problem.
- Rather than relying on historical data directly, we use them to approximate the payouts under a set of economic states.
- The uncertainty is then between these scenarios.

A Toy Case Study: Portfolio Selection

Constructing the Action Space:

- We consider $|\mathcal{G}| = 6$ portfolios constructed from ETF's.
- Datas consists of monthly log-returns over the period 2015–2024 obtained from Yahoo Finance.
- The six portfolios are defined as fixed convex combinations of the underlying ETF assets, representing heterogeneous investment strategies:

	Equity	Bonds	Commodities	Real Estate	Intl. Equity
X_1 (Equity Core)	0.70	0.10	0.05	0.10	0.05
X_2 (Balanced Equity)	0.50	0.20	0.05	0.15	0.10
X_3 (Multi-Asset)	0.30	0.30	0.10	0.20	0.10
X_4 (Bond Dominant)	0.15	0.70	0.05	0.05	0.05
X_5 (Real-Asset Tilt)	0.25	0.15	0.25	0.30	0.05
X_6 (Equal-Weight)	0.20	0.20	0.20	0.20	0.20

A Toy Case Study: Portfolio Selection

Constructing the State Space:

- We define the state space as a set of four economic regimes:

$$S = \{s_1, s_2, s_3, s_4\} = \{\text{Expansion}, \text{Recession}, \text{Stagnation}, \text{Recovery}\}.$$

- The returns under these are approximated using k-means clustering in the joint space of broad market performance and risk.
- Induces partition $\{T_1, \ldots, T_4\}$, with T_j data assigned to state s_j.

A Toy Case Study: Portfolio Selection

Constructing the Utility Function:

The utility of $X \in \mathcal{G}$ in state s_j is defined as the conditional mean return:

$$u \circ X(s_j) := \frac{1}{|T_j|} \sum_{t \in T_j} r_t(X),$$

where $r_t(X)$ is the return of portfolio a at time t.

	Expansion	Recovery	Stagnation	Recession
X_1	0.021	0.059	-0.011	-0.049
X_2	0.018	0.051	-0.011	-0.037
X_3	0.015	0.051	-0.011	-0.029
X_4	0.006	0.024	-0.007	-0.024
X_5	0.014	0.042	-0.008	-0.045
X_6	0.015	0.041	-0.009	-0.024

A Toy Case Study: Portfolio Selection

Constructing the Prior Distributions:

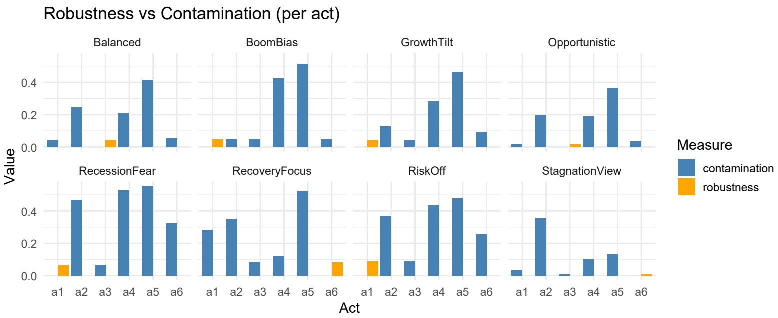
We consider a collection of **eight** prior distributions on the state space S. Each prior $\pi_0 \in \Delta_S$ represents a stylized belief about the likelihood of regimes and is constructed to reflect distinct economic narratives.

Concretely, we consider:

- Four regime-focused priors with largest mass on a single state.
- Four balanced and mixed prior specifications.

A Toy Case Study: Portfolio Selection

Results:



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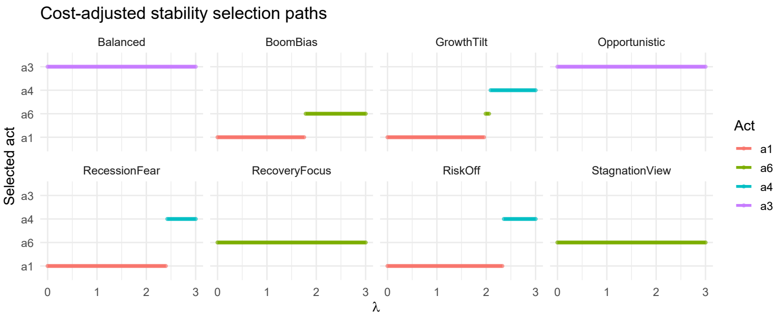

Future Research

- Explore theoretical conncetions between IP, regularisation and stability.
- Other application such as algorithm comparison.

Thank you for your attention!

I am happy to receive your comments and questions.